

GTU ROBOTICS CLUB

Start the using custom docker page:

First Approach:

```
0: 800x800 (no detections), 12.5ms
Speed: 5.5ms preprocess, 12.5ms inference, 0.5ms postprocess per image at shape (1, 3, 800, 800)
root@0694fdbd00f0:/ultralytics# exit
exit
purushottam@purushottam-IdeaPad-Gaming-3-15IHU6:~$ sudo docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
ultralytics_backup   latest          83c8bc1c271    9 days ago     56.1GB
purushottam@purushottam-IdeaPad-Gaming-3-15IHU6:~$ sudo docker ps -a
CONTAINER ID       IMAGE                COMMAND         CREATED        STATUS          PORTS          NAMES
0694fdbd00f0       ultralytics_backup   "/bin/bash"    9 days ago    Exited (0) 11 seconds ago           ultralytics_container
purushottam@purushottam-IdeaPad-Gaming-3-15IHU6:~$ sudo docker commit 0694fdbd00f0 ultralytics_d455:latest
sha256:d77873a5e74c734ce28fdb85a4997ccd568e43b937ba1614594e56cd7d8b61f8
purushottam@purushottam-IdeaPad-Gaming-3-15IHU6:~$ sudo docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
ultralytics_d455     latest          d77873a5e74c    15 seconds ago  62.1GB
ultralytics_backup   latest          83c8bc1c271    9 days ago     56.1GB
```

1.Start with check images and containers

2.run the cmd :

sudo docker commit 0694fdbd00f0 ultralytics_d455:latest

3.check **sudo docker images** cmd

This will create a new your own docker custom image

When I said:

```
sudo docker save -o ultralytics_d455_backup.tar ultralytics_d455:latest
```

- **ultralytics_d455:latest** = the image you just created in **Step 1**.
- **ultralytics_d455_backup.tar** = the portable backup file you want to generate.

This file (.tar) is basically your **entire configured environment** in one package.

Anyone can load it back with:

Cmd: sudo docker load -i ultralytics_d455_backup.tar

Enter into the container with all access on another pc

**Cmd : sudo docker run -it --name ultralytics_container **
**--gpus all **
**--device /dev/bus/usb:/dev/bus/usb **
**--device /dev/video0:/dev/video0 **
**--privileged **
**-e DISPLAY=\$DISPLAY **
**-v /tmp/.X11-unix:/tmp/.X11-unix **
ultralytics_backup:latest /bin/bash

Now it completely plug and play

Second Approach:

1. make the folder in in current container

```
root@0694fbd00f0:/ultralytics# ls
2025-09-23-161706.webm  ai.jpeg  cuda-repo-ubuntu2204-13.0-1-580.82.07-1_and64.deb  nv-tensorrt-local-repo-ubuntu2204-10.13.2-cuda-13.0_1.0-1_and64.deb  trained_model_tensorrt.py
CITATION.cff           best.engine  cudnn-local-repo-ubuntu2204-9.13.0_1.0-1_and64.deb  pyproject.toml  ultralytics
CONTRIBUTING.md       best.onnx   docker                                              realsense_test.py  ultralytics_egg-info
LICENSE                best.pt     docs                                              train.py          ultralytics.engine
README.md              bus.jpg     examples                                           trained_image.py   yolo11n.engine
README.zh-CN.md       convert.py  mkdocs.yml                                         train_model.py     yolo11n.onnx
best.pt                /ultralytics# ls /ultralytics_full
best.pt
root@0694fbd00f0:/ultralytics# cp /ultralytics/best.engine /ultralytics_full/
root@0694fbd00f0:/ultralytics# ls /ultralytics_full
best.engine  best.pt
root@0694fbd00f0:/ultralytics# cp /ultralytics/best.onnx /ultralytics_full/
root@0694fbd00f0:/ultralytics# cp /ultralytics/trained_model_tensorrt.py /ultralytics_full/
root@0694fbd00f0:/ultralytics# ls /ultralytics_full
best.engine  best.onnx  best.pt  trained_model_tensorrt.py
root@0694fbd00f0:/ultralytics# cp /ultralytics/2025-09-23-161706.webm /ultralytics_full/
root@0694fbd00f0:/ultralytics# cp /ultralytics/ai.jpeg /ultralytics_full/
root@0694fbd00f0:/ultralytics# cp /ultralytics/yolo11n.engine /ultralytics_full/
root@0694fbd00f0:/ultralytics# cp /ultralytics/bus.jpg /ultralytics_full/
root@0694fbd00f0:/ultralytics# cp /ultralytics/trained_image /ultralytics_full/
cp: cannot stat '/ultralytics/trained_image': no such file or directory
root@0694fbd00f0:/ultralytics# cp /ultralytics/train_model.py /ultralytics_full/
root@0694fbd00f0:/ultralytics# cp /ultralytics/train_model.py /ultralytics_full/
root@0694fbd00f0:/ultralytics# cp /ultralytics/convert.py /ultralytics_full/
root@0694fbd00f0:/ultralytics# ls /ultralytics_full
2025-09-23-161706.webm  ai.jpeg  best.engine  best.onnx  best.pt  bus.jpg  convert.py  trained_image.py  trained_model.py  trained_model_tensorrt.py  yolo11n.engine
root@0694fbd00f0:/ultralytics#
```

Create a folder inside container for all files you want to share
mkdir -p /ultralytics_full

Copy all Python scripts, trained models, engines, and ONNX files
cp /ultralytics/*.py /ultralytics_full/
cp /ultralytics/*.pt /ultralytics_full/
cp /ultralytics/*.engine /ultralytics_full/
cp /ultralytics/*.onnx /ultralytics_full/

2. New docker image never contain those file who came from Host system or nano
So we have to do it above process it current container.

3.Commit the current image for your own custom image

Cmd: sudo docker commit 0694fdbd00f0 ultralytics_d455:latest

As you show down Enter the container with the access of video,photo,gui,lsusb, x11 etc:

**Cmd:sudo docker run -it --name ultralytics_latest_container **
**--gpus all **
**--device=/dev/bus/usb:/dev/bus/usb **
**--device=/dev/video0:/dev/video0 **
**--privileged **
**-e DISPLAY=\$DISPLAY **
**-v /tmp/.X11-unix:/tmp/.X11-unix **
ultralytics_d455 /bin/bash

4.Show the below image carefully Everything is ready :

```
purushottam@purushottan-IdeaPad-Gaming-3-15IHU6:~$ sudo docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS   NAMES
2dd929a0c713   ultralytics_d455      "/bin/bash"             4 minutes ago   Exited (1) 24 seconds ago           ultralytics_latest_container
0694fdbd00f0   ultralytics_backup    "/bin/bash"             9 days ago     Exited (0) 8 minutes ago           ultralytics_container

purushottam@purushottan-IdeaPad-Gaming-3-15IHU6:~$ sudo docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
ultralytics_d455   latest    19c4704b47fd   7 minutes ago   62.2GB
ultralytics_backup latest    83c8bc1c271    9 days ago     56.1GB

purushottam@purushottan-IdeaPad-Gaming-3-15IHU6:~$ sudo docker start -ai ultralytics_latest_container
root@2dd929a0c713:/ultralytics# ls
2025-09-23-161706.webm  ai.jpeg  cuda-repo-ubuntu2204-13.0-local_13.0.1-580.82.07-1_and64.deb  nv-tensorrt-local-repo-ubuntu2204-10.13.2-cuda-13.0.1-0-1_and64.deb  trained_model_tensorrt.py
CITATION.cff           best.engine  cudnn-local-repo-ubuntu2204-9.13.0-1-0-1_and64.deb             pyproject.toml                                                         ultralytics
CONTRIBUTING.md       best.onnx   docker                                                         realsense_test.py                                                     ultralytics.egg-info
LICENSE                best.pt     docs                                                            tests                                                                    yolov11n.engine
README.md              bus.jpg    examples                                                         trained_image.py                                                        yolov11n.onnx
README.zh-CN.md        convert.py  mkdocs.yml                                                       trained_model.py                                                        yolov11n.pt
root@2dd929a0c713:/ultralytics# exit
exit

purushottam@purushottan-IdeaPad-Gaming-3-15IHU6:~$ sudo docker start -ai ultralytics_container
root@0694fdbd00f0:/ultralytics# ls
2025-09-23-161706.webm  ai.jpeg  cuda-repo-ubuntu2204-13.0-local_13.0.1-580.82.07-1_and64.deb  nv-tensorrt-local-repo-ubuntu2204-10.13.2-cuda-13.0.1-0-1_and64.deb  trained_model_tensorrt.py
CITATION.cff           best.engine  cudnn-local-repo-ubuntu2204-9.13.0-1-0-1_and64.deb             pyproject.toml                                                         ultralytics
CONTRIBUTING.md       best.onnx   docker                                                         realsense_test.py                                                     ultralytics.egg-info
LICENSE                best.pt     docs                                                            tests                                                                    yolov11n.engine
README.md              bus.jpg    examples                                                         trained_image.py                                                        yolov11n.onnx
README.zh-CN.md        convert.py  mkdocs.yml                                                       trained_model.py                                                        yolov11n.pt
root@0694fdbd00f0:/ultralytics#
```

As you checked above our old container has same content as well as in new container also.

5.Start the Testing:

We will do same thing here also

Cmd : python3 trained_model_tensorrt.py(never forget to enter with x11 access)

6. Now this is time to deploy your custom docker image to someone pull and just plug and play

a. Login into the docker or make the new account on docker hub

Link: <https://hub.docker.com/repositories/puru0909>

Then go terminal and run

Cmd: sudo docker login

b. Tag your image:

```
sudo docker tag ultralytics_d455:latest yourdockerhubusername/ultralytics_d455:latest
```

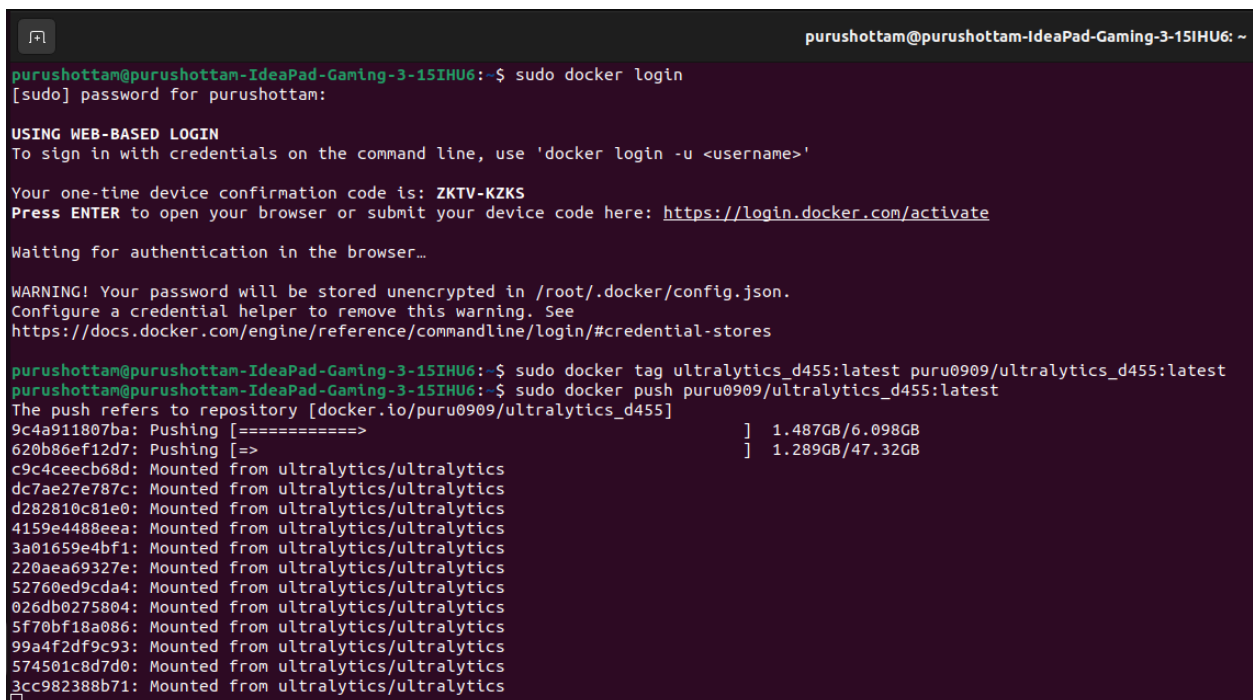
c. Push to Docker Hub:

```
sudo docker push yourdockerhubusername/ultralytics_d455:latest
```

d. Now **anyone** can pull it directly:

```
docker pull yourdockerhubusername/ultralytics_d455:latest
```

(Here my username puru0909)

A terminal window screenshot showing the process of logging into Docker Hub and pushing a custom image. The terminal title is 'purushottam@purushottam-IdeaPad-Gaming-3-15IHU6: ~'. The user runs 'sudo docker login', which prompts for a password. It then shows a 'USING WEB-BASED LOGIN' message with a one-time device confirmation code 'ZKTV-KZKS' and a link to 'https://login.docker.com/activate'. After waiting for authentication, a warning message appears about storing passwords unencrypted. The user then runs 'sudo docker tag ultralytics_d455:latest puru0909/ultralytics_d455:latest' and 'sudo docker push puru0909/ultralytics_d455:latest'. The push process shows progress for two layers: '9c4a911807ba' (1.487GB/6.098GB) and '620b86ef12d7' (1.289GB/47.32GB). A list of 15 layers follows, all marked as 'Mounted from ultralytics/ultralytics'.

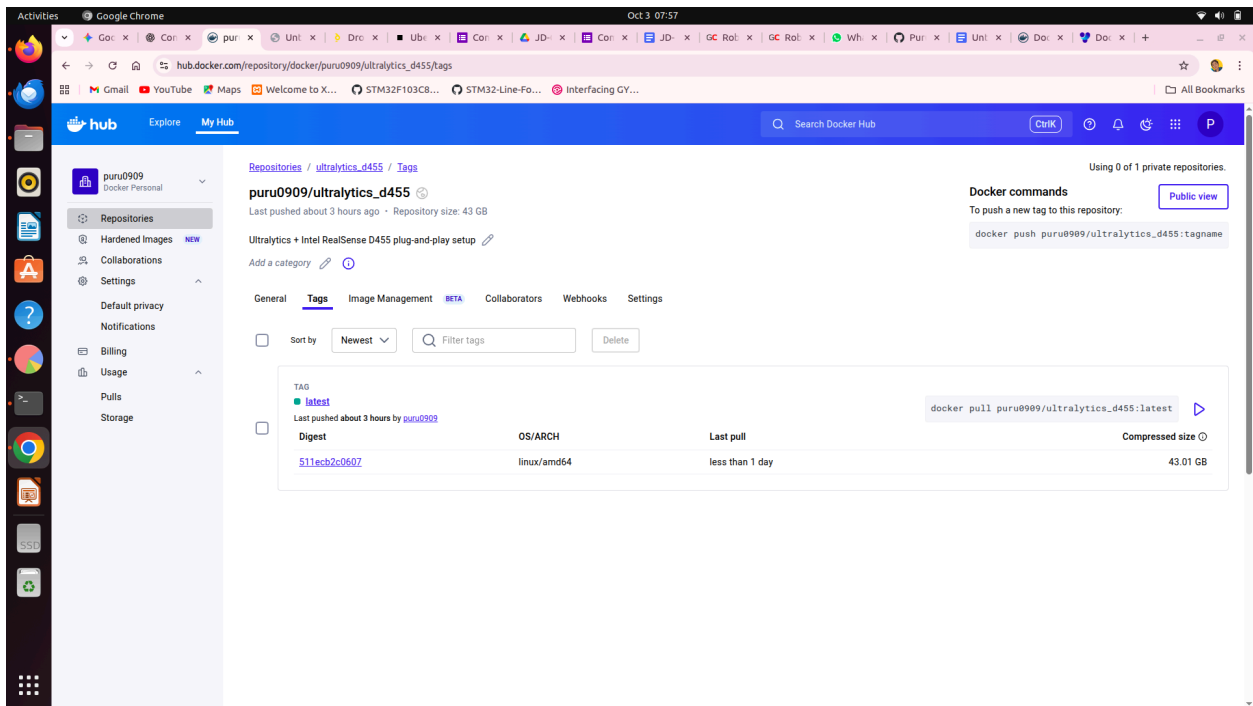
You see something like this during push your own custom image on docker hub

```

purushottam@purushottam-IdeaPad-Gaming-3-15IHU6:~$ sudo docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
ultralitics_d455     latest             19c4704b47fd       16 hours ago       62.2GB
puru0909/ultralitics_d455 latest             19c4704b47fd       16 hours ago       62.2GB
ultralitics_backup   latest             83c8bc1c271        9 days ago         56.1GB
purushottam@purushottam-IdeaPad-Gaming-3-15IHU6:~$ sudo docker images
[sudo] password for purushottam:
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
ultralitics_d455     latest             19c4704b47fd       19 hours ago       62.2GB
puru0909/ultralitics_d455 latest             19c4704b47fd       19 hours ago       62.2GB
ultralitics_backup   latest             83c8bc1c271        9 days ago         56.1GB
purushottam@purushottam-IdeaPad-Gaming-3-15IHU6:~$ sudo docker ps -a
[sudo] password for purushottam:
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS              NAMES
2dd929a0c713       ultralitics_d455    "/bin/bash"        19 hours ago       Exited (0) 18 hours ago           ultralitics_latest_container
0694fdbd00f0       ultralitics_backup  "/bin/bash"        9 days ago         Exited (0) 4 hours ago           ultralitics_container
purushottam@purushottam-IdeaPad-Gaming-3-15IHU6:~$

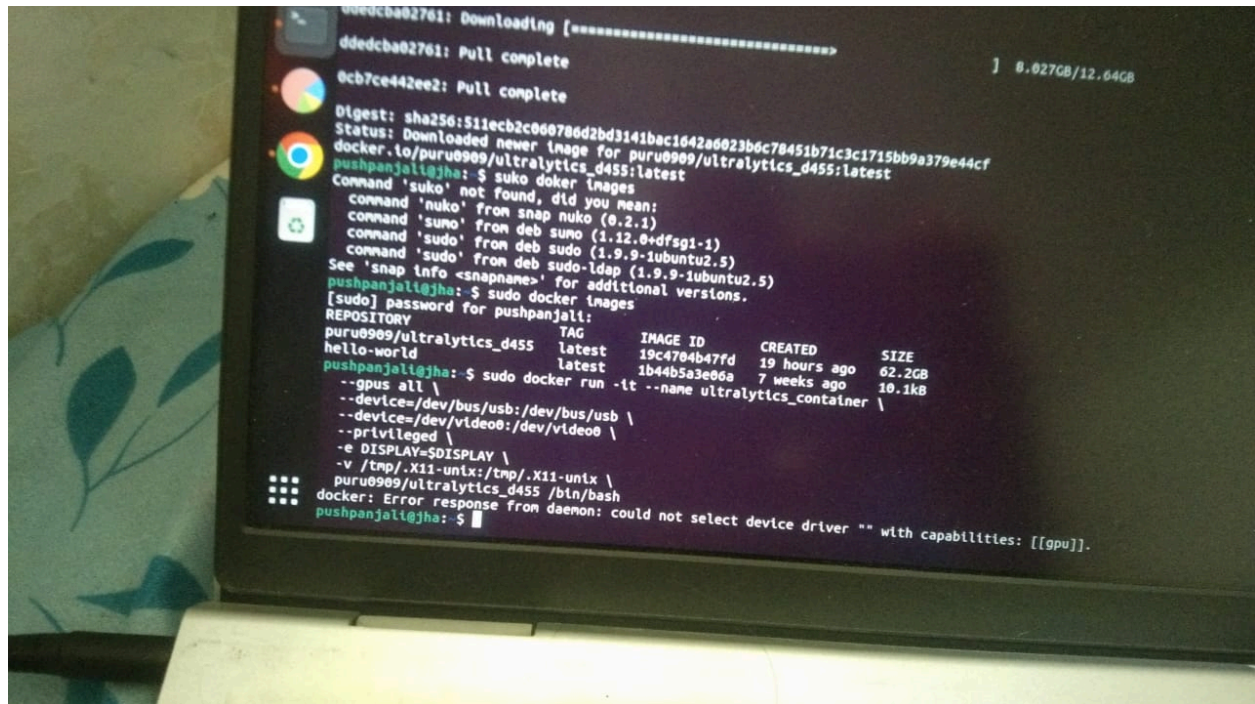
```

Push in done on official website of docker hub



Pull the image on another pc

Cmd: sudo docker pull puru0909/ultralitics_d455:latest



Now everything is ready

Enter the container with all hardware access (photo,video,usb etc)

Just plug and play intel d455 realsense

You see the magic